COSMIN ILIE

Astrophysicist and Cosmologist

Dark matter • First stars and galaxies • Early universe cosmology

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ACADEMIC APPOINTMENTS

Colgate University — Hamilton, NY
Department of Physics and Astronomy
Assistant Professor — 2021–present
Senior Lecturer — 2020–2021
Visiting Assistant Professor — 2018–2020
Lecturer and Research Associate — 2016–2018

University of North Carolina at Chapel Hill — 2014–2016
Postdoctoral Research Associate — Chapel Hill, NC

NASA and CREST Centers at North Carolina Central University — 2013–2014
Postdoctoral Fellow — Durham, NC

National Institute for Physics and Nuclear Engineering — Magurele, Romania
Scientific Researcher, 2013–2020; Postdoctoral Researcher, 2012–2013

Los Alamos National Laboratory — 2011–2012
Visitor, Subatomic Physics Group (P-25) — Los Alamos, NM

University of Michigan — 2008–2011
Research Assistant — Ann Arbor, MI

EDUCATION

University of Michigan — 2011
Ph.D. in Physics — Ann Arbor, MI
Advisor: Katherine Freese | Dissertation: *Selected Topics on the Dark Side*

University of Bucharest — 2001
B.S. in Theoretical Physics — Bucharest, Romania
Thesis: *Path Integrals and Their Applications in Physics*

HONORS AND RESEARCH FUNDING

PRIZES

National Academy of Sciences Cozzarelli Prize — 2023

FELLOWSHIPS

Department of Physics Fellowship, University of Michigan — 2002
Romanian Government Fellowship — 1997–2001

RESEARCH FUNDING

Picker Interdisciplinary Science Institute Major Grant, $84,500; Role: PI; *Probing the Nature of Dark Matter with the First Stars and Galaxies in the Universe* — Apr. 2024–Jun. 2026

New York Space Grant Faculty Fellowship, $10,000; Role: PI; Grant No. 90830-20365; *The first stars in the Universe as Dark Matter probes* — Nov. 2023–Jun. 2024

Rackham Graduate Student Research Grant, University of Michigan — 2011

PUBLICATIONS

REFEREED JOURNAL ARTICLES

Undergraduate coauthors are marked †. Publication titles are linked to the journal or preprint record.

P24 A. A. Siddiqa ('28)†, S. Mahmud ('26)†, and **C. Ilie**. *Neural Network Identification of Dark Star Candidates. II. Spectroscopy*. *Astronomy and Computing* **58**, 101195. 2026

P23 S. Mahmud ('26)†, A. A. Siddiqa ('28)†, and **C. Ilie**. *Neural Network Identification of Dark Star Candidates. I. Photometry*. *Astronomy and Computing* **58**, 101182. 2026

P22 S. Ghodla and **C. Ilie**. *Reconstructing PTA Measurements via Early Seeding of Supermassive Black Holes*. *Physical Review D* **114**, L041303. 2026

P21 **C. Ilie**, J. Paulin ('23)†, A. Petric, and K. Freese. *Supermassive Dark Stars and Their Remnants as a Possible Solution to Three Recent Cosmic Dawn Puzzles*. *Universe* **12**, 1. Featured on the journal cover. 2026

P20 **C. Ilie**, S. Mahmud ('26)†, J. Paulin ('23)†, and K. Freese. *Spectroscopic Supermassive Dark Star Candidates*. *Proceedings of the National Academy of Sciences* **122**, e2513193122. 2025

P19 J. Diks ('25)† and **C. Ilie**. *Constraining Asymmetric Dark Matter Properties by Black Hole Formation in Neutron Stars and Population III Stars*. *Journal of Cosmology and Astroparticle Physics* 09, 017. 2025

P18 R. Casey ('24)† and **C. Ilie**. *Dark Sector Tunneling Field Potentials for a Dark Big Bang*. *Physical Review D* **110**, 103522. 2024

P17 **C. Ilie**. *Closed-Form Expressions for Multiscatter Dark Matter Capture Rates*. *The Astrophysical Journal* **970**, 159. 2024

P16 **C. Ilie**, C. Levy ('23)†, and J. Diks ('25)†. *The Effectiveness of Exoplanets and Brown Dwarfs as Sub-GeV Dark Matter Detectors*. *Journal of Cosmology and Astroparticle Physics* 04, 082. 2024

P15 S. Zhang ('19)†, **C. Ilie**, and K. Freese. *Detectability of Supermassive Dark Stars with the Roman Space Telescope*. *The Astrophysical Journal* **965**, 121. 2024

P14 **C. Ilie**, J. Paulin ('23)†, and K. Freese. *Supermassive Dark Star Candidates Seen by JWST*. *Proceedings of the National Academy of Sciences* **120**, e2305762120. Recipient of the 2023 National Academy of Sciences Cozzarelli Prize. 2023

P13 **C. Ilie** and J. Paulin ('23)†. *Analytic Approximations for the Velocity Suppression of Dark Matter Capture*. *The Astrophysical Journal* **932**, 46. 2022

P12 **C. Ilie** and C. Levy ('23)†. *Multicomponent Multiscatter Capture of Dark Matter*. *Physical Review D* **104**, 083033. 2021

P11 **C. Ilie**, C. Levy ('23)†, J. Pilawa ('20)†, and S. Zhang ('19)†. *Constraining Dark Matter Properties with the First Generation of Stars*. *Physical Review D* **104**, 123031. 2021

P10 **C. Ilie**, J. Pilawa ('20)†, and S. Zhang ('19)†. *Comment on "Multiscatter Stellar Capture of Dark Matter."* *Physical Review D* **102**, 048301. 2020

P9 **C. Ilie** and S. Zhang ('19)†. *Multiscatter Capture of Superheavy Dark Matter by Population III Stars*. *Journal of Cosmology and Astroparticle Physics* 12, 051. 2019

P8 **C. Ilie**. *Gamma-Ray Polarimetry: A New Window for the Non-Thermal Universe*. *Publications of the Astronomical Society of the Pacific* **131**, 111001. 2019

P7 M. Eingorn, L. Fernando, B. Vlahovic, **C. Ilie**, B. Wojtsekhowski, G. M. Urciuoli, F. De Persio, and F. Meddi. *A High-Energy Photon Polarimeter for Astrophysics*. *Journal of Astronomical Telescopes, Instruments, and Systems* **4**(1), 011006. 2018

P6 I. Waldstein, A. Erickcek, and **C. Ilie**. *A Quasi-Decoupled State for Dark Matter in Non-Standard Thermal Histories*. *Physical Review D* **95**, 123531. 2017

P5 B. Vlahovic, M. Eingorn, and **C. Ilie**. *Uniformity of the CMB as a Non-Inflationary Geometrical Effect*. *Modern Physics Letters A* **30**, 1530026. 2015

P4 **C. Ilie**, K. Freese, M. Valluri, I. T. Iliev, and P. Shapiro. *Observing Supermassive Dark Stars with the James Webb Space Telescope*. *Monthly Notices of the Royal Astronomical Society* **422**, 2164. 2012

P3 **C. Ilie**, K. Freese, and D. Spolyar. *Dark Stars and Boosted Dark Matter Annihilation Rates*. *New Journal of Physics* **13**, 053050. 2011

P2 K. Freese, **C. Ilie**, D. Spolyar, M. Valluri, and P. Bodenheimer. *Supermassive Dark Stars: Detectable in JWST*. *The Astrophysical Journal* **716**, 1397–1407. 2010

P1 **C. Ilie**, T. Biswas, and K. Freese. *Are We Seeing the Beginnings of Inflation?* *Physical Review D* **80**, 103521. 2009

OTHER PUBLICATIONS (NON-REFEREED)

N2 **C. Ilie**, C. Levy ('23)†, J. Pilawa ('20)†, and S. Zhang ('19)†. *Probing Below the Neutrino Floor with the First Generation of Stars*. arXiv:2009.11478. Non-refereed letter companion to P11. 2020

N1 K. Freese, E. Ruiz, M. Valluri, **C. Ilie**, D. Spolyar, and P. Bodenheimer. *Supermassive Dark Stars: Detectable in JWST and HST*. *AIP Conference Proceedings* **1294**. 2010

MANUSCRIPTS UNDER REVIEW

S2 **C. Ilie**. *JWST's Little Red Dots as Collapsed Supermassive Dark Stars*. *The Astrophysical Journal Letters* (Revise and Resubmit, revision submitted September 2026). arXiv:2606.02539. 2026

S1 K. Freese, G. M. Fuller, S. Ghodla, **C. Ilie**, K. S. Kehrer, T. Rindler-Daller, and E. I. Sfakianakis. *Early Formation of Supermassive Black Holes via Dark Star Gravitational Instability*. *Proceedings of the National Academy of Sciences* (Revise and Resubmit, revision submitted September 2026). arXiv:2511.08578. 2025

INVITED TALKS

Dark Stars: A Review 2026
Identification of Dark Matter 2026, Zaragoza, Spain

Dark Stars (special evening 1-hour talk) 2026
Lake Louise Winter Institute, Lake Louise, Alberta, Canada

Dark Stars and Their Supermassive Remnants 2025
Space Telescope Science Institute, Galaxies and AGN Seminar

Dark Stars: A Possible Solution to Two Key Astronomical Puzzles 2024
Simons Foundation Center for Computational Astrophysics, New York, NY

Dark Matter and the First Stars (keynote address) 2024
Lake Louise Winter Institute, University of Alberta, Canada

JWST: A New Era for the First Stars and Galaxies as Dark Matter Probes 2024
Dark Matter, First Light workshop, Perimeter Institute, Waterloo, Ontario, Canada

The First Generation of Stars as Dark Matter Probes 2022
XENON Collaboration Seminar

What can the first generation of stars tell us about dark matter? 2020
Colgate University, NASC Colloquium, Hamilton, NY

Cyclic Inflation in Phantom Cosmologies 2012
IFIN-HH DFT Seminar, Magurele, Romania

Dark Stars and Their Detectability 2011
SF11 Cosmology Summer Workshop, Santa Fe, NM

CONTRIBUTED CONFERENCE TALKS IN THE PAST 5 YEARS

Dark Stars: A Possible Solution to Two Recent Astronomical Puzzles 2025
Dark Matter and Stars international conference, Kingston, Ontario, Canada

Dark Stars: A Possible Solution to Two Recent Astronomical Puzzles 2025
Dark Side of the Universe conference, Montreal, Canada

Supermassive Dark Stars and Their Remnants 2024
The Origin and Evolution of Supermassive Black Holes, Sexten, Italy

Probing the Nature of Dark Matter with the First Stars and Galaxies in the Universe 2024
Identification of Dark Matter conference, L'Aquila, Italy

Supermassive Dark Star Candidates in JWST Data 2023
American Astronomical Society 242nd Meeting, Albuquerque, NM

Using the Roman Space Telescope to Find Supermassive Dark Stars 2023
Roman Emerging Science Conference, Space Telescope Science Institute, Baltimore, MD

The First Stars in the Universe as Dark Matter Laboratories 2023
Dark Matter 2023, Santander, Spain

Probing Below the Neutrino Floor with the First Generation of Stars 2022
American Physical Society April Meeting, New York, NY

Population III Stars as Dark Matter Laboratories 2021
Dark Matter 2021, Madrid, Spain (virtual)

Probing Below the Neutrino Floor with the First Generation of Stars 2021
PHENO 2021, Pittsburgh, PA (virtual)

RESEARCH VISIBILITY AND PUBLIC ENGAGEMENT

Selected coverage from more than 200 news stories, interviews, and videos includes:

- *Epsiloon*, "Étoiles noires : le nouvel espoir" (cover story, 2026)
- BBC *Science Focus*, "This radical 'dark stars' theory could solve our Universe's greatest mystery" (2026)
- *New Scientist*, "We may have discovered the first-ever stars powered by dark matter" (2025)
- NPR *Science Friday*, "Have Astrophysicists Spotted Evidence For 'Dark Stars'?" (2025)
- *New Scientist*, "Dark stars: Have we finally found a weird sun powered by dark matter?" (cover feature, 2023)
- Reuters, "Webb telescope captures tantalizing evidence for mysterious 'dark stars'" (2023)
- PBS *Space Time*, "Did JWST Discover Dark Matter Stars?" (2023; 1.5M+ YouTube views)
- *Scientific American*, "JWST Might Have Spotted the First Dark Matter Stars" (2023)
- *El País*, "¿Existen las estrellas oscuras?" (2023)
- *National Geographic France*, "Des astrophysiciens pensent avoir repéré des 'étoiles noires' pour la première fois" (2023)
- *Astronomy*, "Dark stars come into the light" (2018)
- BBC Future, "Did 'dark stars' help form the Universe?" (2016)

TEACHING

COURSES TAUGHT AT COLGATE UNIVERSITY

Core curriculum	CORE S122, *Our Universe: 95% Unknown*	Fall 2026
Introductory	PHYS 105, *Mechanical Physics*	Fall 2024
	PHYS 112, *Fundamental Physics II*	Spring 2019, 2021–2022
	PHYS 131, *Atoms and Waves*	Fall 2016–2017, 2020
Intermediate	PHYS 205, *Mathematical Methods of Physics*	Fall 2022; Spring/Fall 2024
	PHYS 232, *Introduction to Mechanics*	Spring 2020
Advanced	ASTR 414, *Astrophysics*	Spring 2018, 2020, 2022, 2024, 2026
	PHYS 431, *Classical Mechanics*	Fall 2018–2023
	PHYS 432, *Electromagnetism*	Spring 2017
	PHYS 456, *Relativity and Cosmology*	Spring 2019, 2023
Capstone and independent research	PHYS 410, *Advanced Topics and Experiments*	Fall 2016–2020, 2022–2025
	PHYS 491, *Independent Study*	Spring 2017
	PHYS 492 / ASTR 492, *Independent Study – Research*	Spring 2019–2021, 2023–2026

PROFESSIONAL SERVICE

SERVICE TO THE PROFESSION

Proposal Reviewer, NASA FINESST grant program 2025
Mentor, KNAC Summer Research Program 2025
Reviewer, *The Astrophysical Journal* 2022–present
Reviewer, *Journal of Cosmology and Astroparticle Physics* 2024–present
Reviewer, *Astronomy & Astrophysics* 2025–present
Manuscript Reviewer, Princeton University Press 2024–present
Conference Session Chair, Lake Louise Winter Institute 2024, 2026
Session Organizer, *The First Stars and Their Remnants as Dark Matter Probes*, Marcel Grossmann XVII Meeting, Pescara, Italy 2024
Organizer, APS Lilienfeld Prize Public Lecture, Colgate University; speaker: Katherine Freese 2019

INSTITUTIONAL AND DEPARTMENTAL SERVICE

Watson Fellowship Committee	2026–present
Departmental Subcommittee on AI in Teaching	2026–present
Colgate University Research Council	2025–present
Departmental Assessment Coordinator	2024–present
Sigma Pi Sigma Advisor	2024–present
Physics and Astronomy VAP and TS Search Committees	2023–present
Colgate University STARS Program Mentor	2023
Goldwater and Churchill Fellowships Committee	2022–2025
Colgate University Summer Funding Review Committee	2022
Physics and Astronomy Seminar Organizer	2021–2024

UNDERGRADUATE RESEARCH MENTORING

Honors mentees listing research theme, publications we coauthored, awards for research done in the group, and placement.

Manolya Yatman (Bryn Mawr College) *Dark stars* 2025–2026

- *Searching for Supermassive Dark Star Candidates with JWST Photometric Surveys* (ApJ; in prep.).
- Research awards and distinctions: Keck Northeast Astronomy Consortium Research Fellowship.
- Presented our work at several conferences.
- Ph.D. student at the City University of New York.

Sayed Shafaat Mahmud *Dark stars* 2023–2026

- Coauthored three publications: one in *PNAS* and two in *Astronomy and Computing*.
- Research awards and distinctions: Edwin Foster Kingsbury Prize; Phi Beta Kappa Saracino Prize.
- Presented our work at multiple national and international conferences.
- Ph.D. student at Carnegie Mellon University.

Jared Diks *Astrophysical objects as dark matter probes* 2022–2025

- Coauthored two publications in *JCAP*.
- Research awards and distinctions: Amato/Aveni Departmental Award for exceptional research; American Physical Society LeRoy Apker Award nominee.
- Presented our work at multiple national and international conferences.

Lance Chen *Dark stars* 2023–2025

- Presented our work at the APS April Meeting and the NY6 Undergraduate Research Conference.
- M.S. student in Medical Physics at the University of Wisconsin–Madison.

Richard Casey *Early universe cosmology* 2023–2024

- Coauthored a publication in *Physical Review D*.
- Research awards and distinctions: Physics and Astronomy Department Founders Award.
- Presented our work at several international conferences.
- Scientist at SLAC National Accelerator Laboratory.

Jillian Paulin *Dark stars and astrophysical objects as dark matter probes* 2020–2023

- Coauthored four publications: two in *PNAS*, one in *Universe*, and one in *The Astrophysical Journal*.
- Research awards and distinctions: Goldwater Scholar; LeRoy Apker Award national finalist; Amato/Aveni Departmental Award.
- Presented our work at multiple national and international conferences.
- Ph.D. candidate at the University of Pennsylvania.

Caleb Levy *Astrophysical objects as dark matter probes* 2020–2023

- Coauthored four publications: two in *Physical Review D*, one in *JCAP*, and one non-refereed letter.
- Research awards and distinctions: APS Beth Brown Memorial Award; Phi Beta Kappa Saracino Prize.
- Presented our work at multiple national and international conferences.
- Ph.D. student at Harvard University.

Ian Bania *Astrophysical objects as dark matter probes* 2020–2021

- Research awards and distinctions: Amato/Aveni Departmental Award.
- Engineer at IERUS Technologies Inc., Huntsville, Alabama.

Jacob Pilawa *Astrophysical objects as dark matter probes* 2019–2020

- Coauthored three publications: two in *Physical Review D* and one non-refereed letter.
- Research awards and distinctions: Physics and Astronomy Alumni Award; American Physical Society LeRoy Apker Award nominee.
- Earned Ph.D. from the University of California, Berkeley, June 2026.

Saiyang Zhang *Dark stars and astrophysical objects as dark matter probes* 2018–2019

- Coauthored five publications: one in *The Astrophysical Journal*, two in *Physical Review D*, one in *JCAP*, and one non-refereed letter.
- Research awards and distinctions: Amato/Aveni Departmental Award.
- Earned a Ph.D. from the University of Texas at Austin in 2026.
- Postdoctoral fellow at the Canadian Institute for Theoretical Astrophysics, Toronto.

Jonah Kudler-Flam *Early universe cosmology* 2016–2017

- Research awards and distinctions: Phi Beta Kappa Saracino Prize; American Physical Society LeRoy Apker Award nominee.
- Earned a Ph.D. from the University of Chicago in 2022.
- Marvin L. Goldberger Member at the Institute for Advanced Study, Princeton.